Exercise Sheet 7

MATH 598 / 784 · High-Dimensional Probability

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Throughout, $C, c > 0$ denote absolute constants, whose values may change. For a nonempty bounded set T in a Euclidean space, write $w(T) = \mathbb{E}\sup_{x\in T}\langle g, x\rangle$, where g is standard Gaussian.

1. Gaussian width and Gaussian complexity (Exercise 7.6.9)

The *Gaussian complexity* of $T \subset \mathbb{R}^n$ is

$$\gamma(T) = \mathbb{E}\sup_{x\in T}|\langle g, x\rangle|, \qquad g \sim N(0, I_n).$$

1. Let T be nonempty and bounded, and let $y \in T$. Show that
$$\frac{1}{3}\big[w(T) + \|y\|_2\big] \le \gamma(T) \le 2\big[w(T) + \|y\|_2\big].$$
Other absolute constants in place of $1/3$ and 2 are acceptable.
2. Deduce that $w(T) \le \gamma(T) \le 2w(T)$ whenever $0 \in T$.
3. Show that $\gamma(T) = w(T \cup (-T))$. Give an example explaining why $\gamma(T)$ cannot be bounded by a constant multiple of $w(T)$ for arbitrary T.

Hint. First consider a set containing 0, so that the suprema of $\langle g, x\rangle$ and $-\langle g, x\rangle$ are both nonnegative. Then translate by y. Recall that $\mathbb{E}|N(0,1)| = \sqrt{2/\pi}$.

2. A semidefinite relaxation for positive semidefinite quadratic forms

Let A be a real symmetric positive semidefinite $n \times n$ matrix. Define

$$\mathrm{INT}(A) = \max_{s\in\{-1,1\}^n} s^{\mathsf{T}} A s,$$
$$\mathrm{SDP}(A) = \max\{\mathrm{tr}(AX) : X \succeq 0,\ X_{ii} = 1 \text{ for all } i\}.$$

Here $X \succeq 0$ means that X is symmetric positive semidefinite.

1. Show that a feasible X is a Gram matrix $X_{ij} = \langle u_i, u_j\rangle$ of unit vectors $u_1, \dots, u_n \in \mathbb{R}^n$, and that $\mathrm{INT}(A) \le \mathrm{SDP}(A)$.
2. For $g \sim N(0, I_n)$ and a unit vector u, prove
$$\mathbb{E}\big[\operatorname{sgn}(\langle g, u\rangle) g\big] = \sqrt{2/\pi}\, u.$$
3. Given a feasible X and its Gram vectors, set $s_i = \operatorname{sgn}(\langle g, u_i\rangle)$ and $r_i = s_i - \sqrt{2/\pi}\langle g, u_i\rangle$. Show that
$$\mathbb{E}[ss^{\mathsf{T}}] - \frac{2}{\pi}X = \mathbb{E}[rr^{\mathsf{T}}] \succeq 0.$$
4. Deduce that this Gaussian rounding satisfies $\mathbb{E}[s^{\mathsf{T}}As] \ge (2/\pi)\,\mathrm{tr}(AX)$, and hence
$$\frac{2}{\pi}\,\mathrm{SDP}(A) \le \mathrm{INT}(A) \le \mathrm{SDP}(A).$$
Where is the assumption $A \succeq 0$ used?

Hint. In part 2, decompose g into its component along u and its orthogonal component. For part 4, show that $\mathrm{tr}(AB) \ge 0$ whenever $A, B \succeq 0$. The entries of A need not be nonnegative.

For Problems 3 and 4, let $N(T, \varepsilon)$ be the least number of Euclidean balls of radius ε, with centers in T, covering T. You may use *Dudley's inequality* in the form

$$w(T) \le C \int_0^\infty \sqrt{\log N(T, \varepsilon)}\, d\varepsilon.$$

For matrix spaces, Euclidean distance means Frobenius distance.

3. Dudley's inequality for low-rank matrices

Let $1 \le r \le \min\{m, n\}$ and

$$T_r = \{M \in \mathbb{R}^{m \times n} : \operatorname{rank}(M) \le r,\ \|M\|_F \le 1\}.$$

Let G be an $m \times n$ matrix with independent $N(0,1)$ entries, so that $w(T_r) = \mathbb{E} \sup_{M \in T_r} \operatorname{tr}(G^{\mathsf{T}} M)$.

1. Using the singular value decomposition, prove that, for $0 < \varepsilon \le 1$,

$$N(T_r, \varepsilon) \le \left(\frac{7}{\varepsilon}\right)^{(m+n+1)r}.$$

2. Apply Dudley's inequality to deduce

$$w(T_r) \le C\sqrt{r(m+n)}.$$

3. Prove the matching lower bound $w(T_r) \ge c\sqrt{r(m+n)}$ by finding a large Euclidean ball contained in T_r. What does the answer give for $r = 1$ and for $r = \min\{m, n\}$?

Hint for part 1. You may use the volumetric fact that any subset of the unit ball of a d-dimensional normed space has a δ-net, with centers in that subset, of size at most $(1 + 2/\delta)^d$. Write $M = U\Sigma V^{\mathsf{T}}$, where U, V have r orthonormal columns and Σ is diagonal, nonnegative, with $\|\Sigma\|_F \le 1$. Cover the sets of U and V in **operator norm**, and the singular-value vectors in Euclidean norm, at scale $\delta = \varepsilon/3$. Use $\|BC\|_F \le \|B\|_{\mathrm{op}} \|C\|_F$ to control the error.

4. When Dudley's bound loses a factor (optional)

For $n \ge 2$, consider the ellipsoid

$$E_n = \left\{ x \in \mathbb{R}^n : \sum_{j=1}^n j x_j^2 \le 1 \right\}.$$

1. Show that

$$w(E_n) = \mathbb{E}\left(\sum_{j=1}^n \frac{g_j^2}{j}\right)^{1/2}, \qquad c\sqrt{\log n} \le w(E_n) \le C\sqrt{\log n}.$$

2. For each $1 \le k \le n$, the first k coordinate directions contain a ball of radius $k^{-1/2}$ lying in E_n. Use projection and volume comparison to prove

$$N(E_n, \varepsilon) \ge \left(\frac{1}{\varepsilon\sqrt{k}}\right)^k.$$

3. Choose k as a function of ε to show that

$$\log N(E_n, \varepsilon) \ge c\varepsilon^{-2} \quad \text{for } n^{-1/2} \le \varepsilon \le \tfrac{1}{2}.$$

   Deduce that $\int_0^\infty \sqrt{\log N(E_n, \varepsilon)}\, d\varepsilon \ge c \log n$. Compare this with the actual Gaussian width.

Hint for part 1. Apply Jensen's inequality to the Euclidean norm of $(|g_j|/\sqrt{j})_{j=1}^n$ for the lower bound, and Cauchy–Schwarz for the upper bound.

Problem 1 is adapted from Exercise 7.6.9 of Roman Vershynin, *High-Dimensional Probability*, first edition (author's PDF). Problems 2–4 develop Gaussian rounding and the entropy integral from the lectures.